CLINOMA PLATFORM

MATCHING ASSESSMENT

Day 2 Pediatrics Quiz Answers Key

Interactive Diagnosis & Classification

Prepared by: Clinoma Platform Team

Set 1: Childhood Obesity (Differential Diagnosis)

Column I (Questions)	[Match]	Column II (Answers)
1. Nutritional Obesity	[B]	A. Decreased/decelerated linear growth. Test for thyroid hormones.
2. Endocrine Obesity	[A]	B. Consistent/accelerated growth, early puberty, advanced bone age (> 2 SD).
3. Genetic Syndrome Obesity	[C]	C. Severe obesity < 5 years. Developmental delay, short stature, dysmorphic facies, hyperphagia.

Set 2: Congenital Hypothyroidism (Causes & Radiographic Features)

Column I (Questions)	[Match]	Column II (Answers)
1. Defect of fetal thyroid development (DYSGENESIS)	[D]	A. Missing or too small gland.
2. Defect in thyroid hormone synthesis (DYSHORMONOGENESIS)	[C]	B. Absent/small epiphyses, stippled appearance, cortical thickening.
3. Thyroid scan	[A]	C. Defect in thyroid hormone synthesis.
4. Bone age (Greulich & Pyle method)	[B]	D. Aplasia, Hypoplasia, Ectopia.

Set 3: Diabetes Mellitus (Insulin Therapy Management)

Column I (Questions)	[Match]	Column II (Answers)
1. Rapid/Short-acting (Lispro, Aspart)	[B]	A. 30–60 mins before meals.
2. Regular	[A]	B. 15 mins before meals.
3. Intermediate (NPH, Lente)	[C]	C. 10–16 hours.
4. Long-acting (Glargine, Detemir)	[D]	D. 20–24 hours.

Set 4: Short Stature (Normal Variants of Growth)

Column I (Questions)	[Match]	Column II (Answers)
1. Familial Short Stature	[A]	A. Bone Age = Chronologic Age (Crucial differential point).
2. Constitutional Delay of Growth and Puberty (CDGP)	[C]	B. Most achieve catch-up growth by 2 years of age to enter the normal range.
3. Idiopathic Short Stature (ISS)	[D]	C. Normal birth size ▯ downward shift at 3–6 months ▯ grows parallel to but below the 3rd percentile by 3–4 years.
4. Small for Gestational Age (SGA) Infants	[B]	D. Height ≤ 2 SD below the mean with no endocrine, metabolic, or systemic diagnosis.

Set 5: Hematological Hallmarks & Pathognomonic Features

Column I (Questions)	[Match]	Column II (Answers)
1. G6PD-deficient individuals	[F]	A. Eosin-5-maleimide (EMA) is the diagnostic test of choice.
2. Hereditary Spherocytosis (HS)	[A]	B. Hypocellular BM for age plus two of the following three criteria: Platelet count < 20,000/mm^3, Absolute reticulocyte count < 40,000/mm^3, or Absolute neutrophil count (ANC) < 500/mm^3.
3. Hodgkin lymphoma (HL)	[C]	C. The Reed-Sternberg (RS) cell is the hallmark diagnostic feature, characterized as a large cell (15-45 μm in diameter) containing multiple or multilobulated nuclei ("owl-eye" appearance).
4. Beta-Thalassemia Major	[E]	D. Bone marrow aspiration and biopsy are confirmative and diagnostic if more than 25% of the bone marrow cells are lymphoblasts.
5. SEVERE APLASTIC ANEMIA CRITERIA	[B]	E. Peripheral smear: TARGET CELLS, ANISOCYTOSIS, POIKILOCYTOSIS.
6. Acute Lymphoblastic Leukemia (ALL)	[D]	F. Denatured hemoglobin aggregates are VISUALIZED as HEINZ BODIES in peripheral blood smears.

Set 6: Genetics & Inheritance Patterns

Column I (Questions)	[Match]	Column II (Answers)
1. Hemophilia A (Classic)	[E]	A. MOST COMMON ENZYMATIC DISORDER of RBCs, inherited via SEX-LINKED RECESSIVE MODE on the X CHROMOSOME.
2. Hemophilia C	[B]	B. Factor XI deficiency. (Autosomal recessive, rarest)
3. HYDROPS FETALIS	[F]	C. Fanconi anemia and Diamond Blackfan syndrome.
4. Hereditary Spherocytosis (HS)	[D]	D. Mode of inheritance: Autosomal Dominant (AD) in 75% of cases.
5. G6PD DEFICIENCY	[A]	E. Factor VIII deficiency. (X-linked recessive, affects males)
6. CONGENITAL (Mostly Inherited) Etiology of Aplastic Anemia	[C]	F. SEVERE HEMOLYSIS in utero, leading to DEATH (4 genes deleted).

Set 7: Clinical Manifestations & Unique Signs

Column I (Questions)	[Match]	Column II (Answers)
1. Iron deficiency anemia (IDA)	[C]	A. Nonspecific signs including anorexia, fatigue, malaise, irritability, and intermittent low-grade fever.
2. IMMUNE THROMBOCYTOPENIA (ITP)	[B]	B. SUDDEN APPEARANCE of PETECHIAL RASH, PURPURA, & ECCHYMOSES (BRUISING) in an otherwise healthy child.
3. BETA-THALASSEMIA MAJOR (COOLEY ANEMIA)	[D]	C. pica, which is the intense craving for nonfood items like clay, dirt, rocks, starch, chalk, soap, paper, or cardboard.
4. Childhood Acute Lymphoblastic Leukemia (ALL) Initial presentation	[A]	D. "THALASSEMIC FACIES" (frontal bossing, flat nasal bridge, maxilla hyperplasia)
5. Hemophilia in Mobile Children & Adolescents	[E]	E. Hemarthrosis (Joint Bleeding): Most common site (up to 80% of hemorrhages). Affects ankles, knees, elbows.

Set 8: Laboratory Profiles & Investigations

Column I (Questions)	[Match]	Column II (Answers)
1. Iron profile in IDA	[E]	A. INCREASED indirect bilirubin, high serum ferritin, high serum iron, high transferrin saturation.
2. Diagnosis in ITP	[C]	B. Coagulation time & Activated partial thromboplastin time (aPTT): Prolonged
3. Blood chemistry in ALL	[D]	C. CBC shows ISOLATED THROMBOCYTOPENIA
4. Biochemistry in Beta-Thalassemia Major	[A]	D. shows markedly elevated lactate dehydrogenase (LDH) and uric acid.
5. Investigations in Hemophilia	[B]	E. shows low serum iron, low serum ferritin, increased Total iron binding capacity (TIBC), and decreased transferrin saturation (TS).

Set 9: Severe Complications, Crises & Emergencies

Column I (Questions)	[Match]	Column II (Answers)
1. APLASTIC CRISIS	[B]	A. IgG antibodies that neutralize clotting factor concentrates. Occurs in 30% of severe Hemophilia A.
2. MEGALOBLASTIC CRISIS	[C]	B. MOST DANGEROUS! Almost universally triggered by Parvovirus B19. Completely halts marrow erythropoiesis.
3. HEMOLYTIC CRISIS	[D]	C. Due to rapid, severe depletion of nutritional folate stores driven by chronic, highly accelerated bone marrow erythropoiesis.
4. Tumor Lysis Syndrome (TLS)	[E]	D. Sudden, temporary acceleration of RBC destruction, precipitated by acute bacterial or viral infections.
5. Inhibitors (Major Complication)	[A]	E. A life-threatening complication managed with aggressive IV hydration before chemotherapy, uric acid reduction via allopurinol, and close electrolyte monitoring.

QUIZ ANSWER KEY (□□□□□□ □□□□□□ □□□□□□□)

Set 1: 1 □ B, 2 □ A, 3 □ C

Set 2: 1 □ D, 2 □ C, 3 □ A, 4 □ B

Set 3: 1 □ B, 2 □ A, 3 □ C, 4 □ D

Set 4: 1 □ A, 2 □ C, 3 □ D, 4 □ B

Set 5: 1 □ F, 2 □ A, 3 □ C, 4 □ E, 5 □ B, 6 □ D

Set 6: 1 □ E, 2 □ B, 3 □ F, 4 □ D, 5 □ A, 6 □ C

Set 7: 1 □ C, 2 □ B, 3 □ D, 4 □ A, 5 □ E

Set 8: 1 □ E, 2 □ C, 3 □ D, 4 □ A, 5 □ B

Set 9: 1 □ B, 2 □ C, 3 □ D, 4 □ E, 5 □ A